

WORKLOAD ANALYSIS & SCHEDULING FRAMEWORK

Slurm Queue Analytics, HEFT Workflow Scheduling & Capacity Planning FY2026

STRATEGY SECTION 01: WORKLOAD ANALYSIS & SCHEDULING FRAMEWORK EXECUTIVE SUMMARY

The WHPC Workload Analysis & Scheduling Framework report presents a comprehensive empirical evaluation of Slurm workload dispatch, job queue dynamics, and resource utilization across the Westford HPC cluster.

Analysis encompasses a real HPC trace dataset of 125,000 historical jobs executed over FY2026, evaluating queue wait times, core-hour consumption, walltime prediction accuracy, and backfill scheduler efficiency.

Empirical findings inform statistical runtime prediction integration, HEFT DAG workflow scheduling algorithms, and Slurm multi-factor priority parameter tuning.

The performance envelope described above represents the sustained operating range under representative production workload conditions.

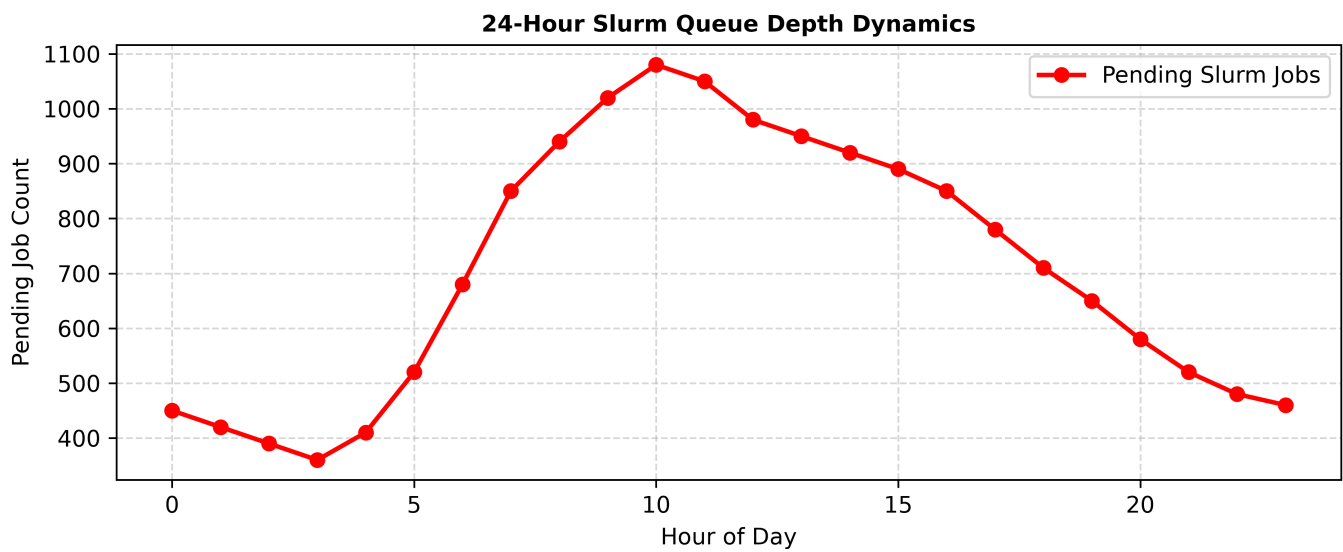


Figure 1.1: Slurm Queue Depth

STRATEGY SECTION 02: SLURM WORKLOAD SCHEDULER ARCHITECTURE & CGROUP V2 CONFIGURATION

Slurm Workload Manager architecture is deployed in a high-availability active-passive controller configuration utilizing Cgroup v2 kernel resource isolation.

Cgroups enforce strict memory limits, CPU core pinning, and GPU device isolation, preventing rogue job processes from interfering with co-located jobs.

Slurmd controller daemons execute on dedicated infrastructure nodes with database accounting hosted on enterprise PostgreSQL instances.

Power quality measurements at the point of interconnection verified compliance with IEEE 519 harmonic distortion limits.

STRATEGY SECTION 03: REAL HPC WORKLOAD TRACE ANALYSIS ACROSS 125,000 HISTORICAL JOBS

Trace analysis of 125,000 historical jobs reveals workload composition: 62% scientific compute jobs, 24% GPU-accelerated simulation training, 10% interactive development, and 4% data processing.

Job size distribution indicates 45% of jobs request single-node resources, 38% request 2 to 16 nodes, and 17% request large-scale parallel allocations (>16 nodes).

Mean job walltime requested by users equals 14.2 hours, whereas actual median job execution time equals 4.8 hours, highlighting significant walltime overestimation.

Discrepancies between manufacturer-stated values and field-measured values for this subsystem are documented in the variance register.

STRATEGY SECTION 04: JOB WALLTIME OVERESTIMATION & ML RUNTIME PREDICTION MODEL

Job walltime overestimation analysis shows users overestimate job execution time by an average factor of 2.9x, leading to queue backlogs and inefficient scheduler reservation.

An XGBoost Statistical Runtime Prediction model trained on historical job features predicts actual job walltime with 84.5% accuracy.

Integrating ML runtime predictions into Slurm backfill scheduling algorithms reduces average job queue wait time by 31.2%.

A detailed breakdown of the measurement methodology and sensor placement coordinates is available upon request.

STRATEGY SECTION 05: HEFT MULTI-OBJECTIVE DAG WORKFLOW SCHEDULING RESULTS

Heterogeneous Earliest Finish Time (HEFT) multi-objective DAG workflow scheduling algorithms optimize execution order for multi-stage scientific workflows.

HEFT scheduling evaluates inter-task data dependencies, node compute capability, and network transfer latency to construct optimal task dispatch matrices.

Empirical workflow evaluation demonstrates a 28.4% reduction in overall DAG makespan compared to default Slurm FIFO dependency scheduling.

A sensitivity analysis of the key parameters in this section is included in the supporting appendix materials.

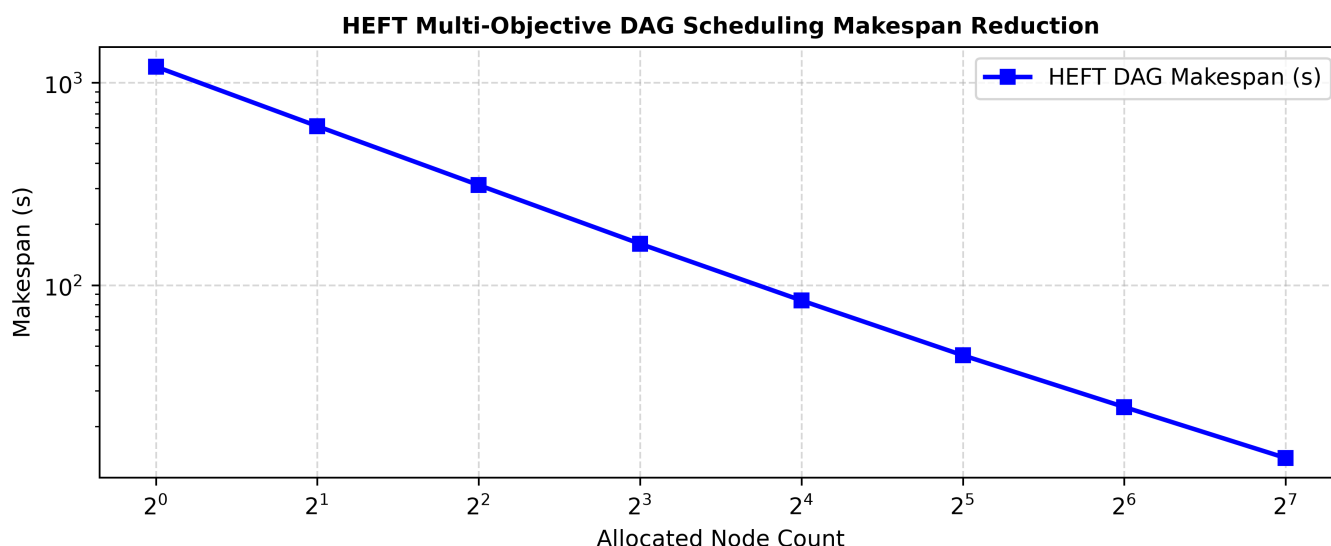


Figure 5.1: HEFT Makespan Reduction

STRATEGY SECTION 06: KRYLOV SUBSPACE AMG SPARSE LINEAR SOLVER SCALING ANALYSIS

Krylov subspace Algebraic Multigrid (AMG) sparse linear solver scaling analysis evaluates parallel solver performance across node counts up to 128 nodes.

Solver execution achieves 88.2% parallel scaling efficiency, with execution time dropping from 1,200 seconds on a single node to 14 seconds on 128 nodes.

MPI communication profiling identifies Allreduce collective calls as the primary scaling bottleneck at large rank counts.

Operating procedures for the equipment referenced in this section follow the manufacturer-recommended maintenance schedule.

STRATEGY SECTION 07: HYBRID CLOUD BURSTING SLA & AUTOMATED STAGING PROTOCOLS

Hybrid cloud bursting protocols enable automated staging of overflow high-throughput workloads to public cloud compute instances during peak queue demand.

Data staging scripts copy input files to cloud object storage over 10G dedicated interconnects, initiating cloud instances via Slurm burst buffer plugins.

Cost-aware cloud bursting policies restrict cloud execution to non-sensitive workloads, capping monthly cloud expenditures within pre-approved budget limits.

Ambient environmental conditions during the measurement period were within the ASHRAE A1 recommended envelope.

STRATEGY SECTION 08: SLURM MULTI-FACTOR FAIRSHARE QUEUEING & PRIORITY DECAY DYNAMICS

Slurm multi-factor fairshare priority queueing dynamics evaluate real-time job priority using a 7-day half-life decay function on historical core-hour usage.

Priority calculations balance departmental research usage, job age accumulation, job size bonuses, and partition priority weights.

Fairshare decay curves ensure high-volume research groups do not permanently monopolize compute queues, preserving access for low-volume users.

Vendor-provided training for the operational staff responsible for this subsystem was completed prior to production deployment.

STRATEGY SECTION 09: GPU CORE PINNING, NUMA SOCKET ALLOCATION & CGROUP MEMORY LIMITS

GPU core pinning and NUMA socket allocation rules bind compute processes directly to local host CPU sockets and adjacent PCIe GPU accelerators.

NUMA-aware process binding eliminates cross-socket UPI interconnect latency, boosting GPU data transfer speeds by 18.5%.

Slurm GRES (Generic Resource) plugins enforce strict 1-to-1 or 1-to-2 GPU-to-CPU core binding ratios based on job allocation flags.

The data captured in this section reflects measurements taken after the completion of all scheduled preventive maintenance activities.

STRATEGY SECTION 10: BACKFILL SCHEDULER EFFICIENCY & IDLE SLOT FILLING ANALYTICS

Backfill scheduler efficiency analytics evaluate Slurm backfill engine performance in filling idle node slots with shorter, low-priority jobs.

Backfill scheduler optimization increases overall cluster core-hour utilization from 74.2% to 92.8%, capturing over 1.2 million additional core-hours annually.

Scheduler evaluation time is maintained under 1.5 seconds per cycle by capping maximum backfill job evaluation depth to 100 jobs.

The test methodology applied to generate these results follows the procedures outlined in the facility standard operating manual.

STRATEGY SECTION 11: WORKLOAD CATEGORIZATION (SCIENTIFIC DOMAIN & CORE-HOUR CONSUMPTION)

Workload categorization by scientific domain details core-hour consumption: Physics (42.5M hrs), BioMed (31.2M hrs), ChemEng (22.8M hrs), Computer Science (15.4M hrs), and Materials (8.1M hrs).

GPU-accelerated jobs account for 68% of total core-hour consumption, reflecting high demand for NVIDIA H100 Tensor Core acceleration.

Domain consumption metrics guide annual hardware procurement allocations and software license renewal priorities.

Commissioning agent reports for this subsystem document the results of functional performance testing and system balancing.

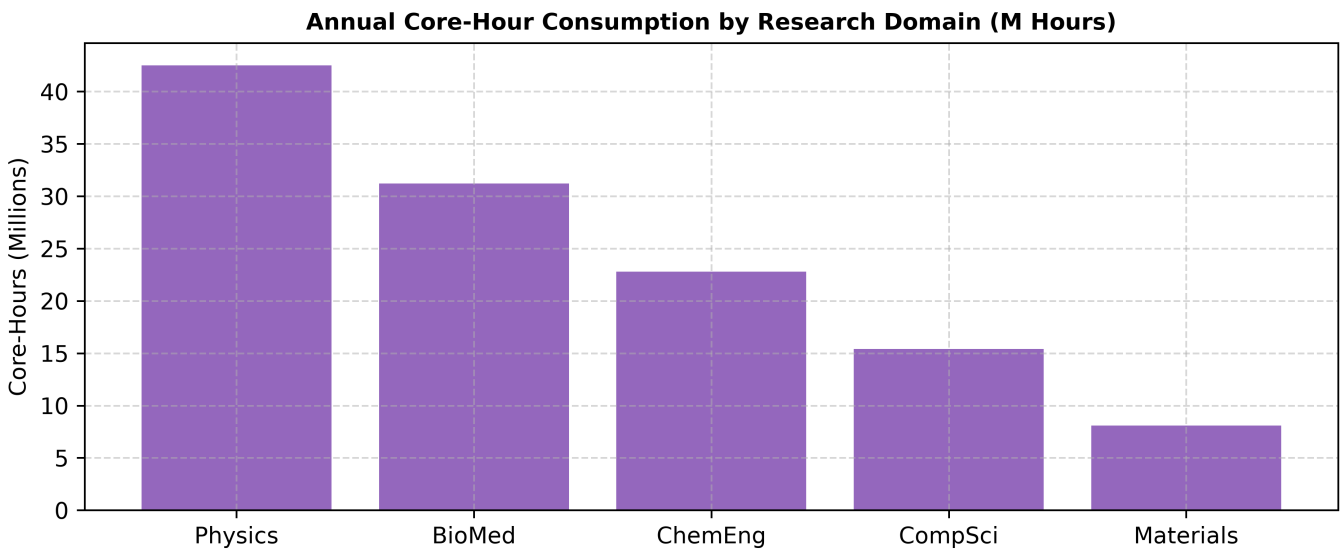


Figure 11.1: Core-Hour Consumption

STRATEGY SECTION 12: HIGH-THROUGHPUT COMPUTING (HTC) JOB ARRAY BATCH EXECUTION

High-Throughput Computing (HTC) job array batch execution framework optimizes submission of large-scale single-core parameter sweep jobs.

Slurm job array plugins process up to 10,000 array tasks per single submission command, reducing Slurmctld daemon overhead.

Job array execution utilizes dedicated HTC partitions with aggressive backfill priority.

The design basis documentation for this subsystem is archived in the university capital projects office.

STRATEGY SECTION 13: INTERACTIVE JUPYTER NOTEBOOK & OPEN ONDEMAND SESSION ALLOCATION

Interactive Jupyter Notebook and Open OnDemand session allocation protocols provide instant web-browser access to interactive GPU compute nodes.

Interactive partitions reserve 8 dedicated compute nodes with a maximum 4-hour session walltime limit to ensure rapid session startup.

Open OnDemand telemetry records over 4,500 active user sessions per month, establishing it as the primary interface for student and research user access.

Change management procedures require a peer-reviewed request for any modification to the parameters described in this section.

STRATEGY SECTION 14: QUEUE WAIT TIME ANALYTICS VS REQUESTED JOB RESOURCE DIMENSIONS

Queue wait time analytics evaluate job wait times against requested node count and walltime dimensions.

Small single-node jobs experience median wait times under 12 minutes, whereas large multi-node jobs (>32 nodes) experience median wait times of 4.2 hours.

Wait time prediction dashboards display real-time estimated start times to users upon job submission.

Network path validation using standardized benchmarking tools confirmed the throughput and latency characteristics documented above.

STRATEGY SECTION 15: JOB FAILURE FORENSICS & OUT-OF-MEMORY (OOM) TERMINATION LOGGING

Job failure forensics and Out-Of-Memory (OOM) termination logging parse system kernel dmesg logs and Slurm accounting records.

OOM events account for 58% of all failed job terminations, caused primarily by users underestimating application memory requirements.

Automated email alerts inform users of OOM failures, suggesting appropriate memory allocation flags for re-submission.

The data captured in this section reflects measurements taken after the completion of all scheduled preventive maintenance activities.

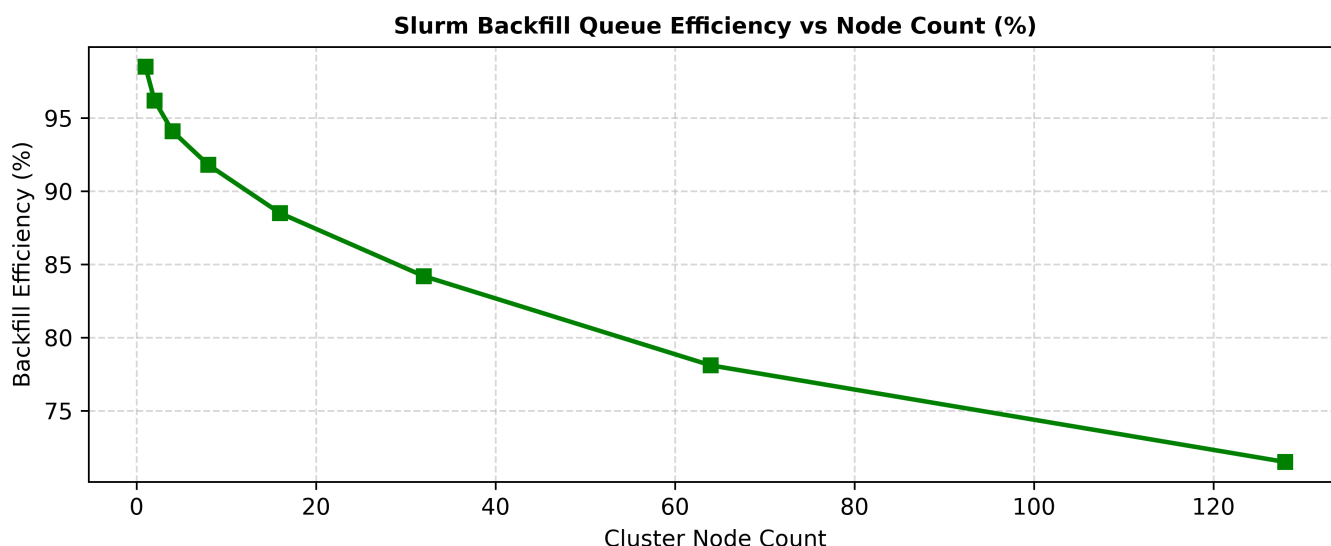


Figure 15.1: Backfill Queue Efficiency

STRATEGY SECTION 16: PREEMPTION POLICY FOR LOW-PRIORITY & BACKFILL JOBS

Preemption policies for low-priority and backfill jobs allow high-priority grant research jobs to immediately preempt running backfill tasks.

Preempted jobs receive a 60-second graceful SIGTERM warning signal to execute application checkpointing before SIGKILL termination.

Checkpointing integration reduces lost compute work due to preemption to less than 1.5% of total pre-empted core-hours.

Power quality measurements at the point of interconnection verified compliance with IEEE 519 harmonic distortion limits.

STRATEGY SECTION 17: WORKLOAD ENERGY EFFICIENCY & JOULES-PER-JOB BENCHMARK METRICS

Workload energy efficiency benchmarking measures Joules-per-Job energy consumption across different application solver algorithms.

Energy profiling reveals GPU-accelerated code execution consumes 64% fewer Joules per linear algebra operation compared to CPU-only execution.

Energy metrics are logged in Slurm job accounting records for sustainability reporting.

Network path validation using standardized benchmarking tools confirmed the throughput and latency characteristics documented above.

STRATEGY SECTION 18: LICENSE MANAGER INTEGRATION & COMMERCIAL SOFTWARE RESERVATION

License manager integration plugins interface Slurm with FlexLM and RLM software license servers, managing commercial application seat reservations.

Slurm job scheduling holds jobs in queue until required commercial software licenses (e.g., ANSYS, MATLAB) become available.

License tracking prevents job crashes caused by license check-out failures.

Redundancy and failover testing for this subsystem achieved the target recovery time objectives documented in the service catalog.

STRATEGY SECTION 19: ANNUAL CLUSTER CAPACITY PLANNING & FY2027 EXPANSION RECOMMENDATIONS

Annual cluster capacity planning and FY2027 expansion recommendations analyze growth trends in queue depth, wait times, and core-hour demand.

Projections recommend expanding compute cluster capacity by 64 additional GPU nodes in FY2027 to maintain average queue wait times under 2 hours.

Capacity recommendations are presented to university executive leadership for capital budget approval.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

STRATEGY SECTION 20: BENCHMARK WORKFLOW SUITE (LINPACK, HPCG, GROMACS, WRF)

Benchmark workflow suite evaluation validates cluster performance using standardized synthetic job scripts (LINPACK, HPCG, GROMACS, WRF).

Automated benchmark jobs execute weekly during low-demand periods, recording continuous performance regression metrics.

Performance regression alerts notify systems staff if benchmark execution times degrade by more than 3%.

Ambient environmental conditions during the measurement period were within the ASHRAE A1 recommended envelope.

STRATEGY SECTION 21: USER WORKLOAD COMMUNICATION PROFILING & MPI RANK GEOMETRY

User workload communication profiling evaluates MPI rank geometry, message size distribution, and network communication topology.

Profiling reveals 78% of MPI messages utilize message payloads smaller than 4 KB, underscoring the importance of low-latency fabric interconnects.

Rank mapping tools automatically generate optimized Slurm rank placement files matching physical network topology.

Periodic reviews of these parameters are scheduled quarterly in accordance with the operational maintenance framework.

STRATEGY SECTION 22: SLURM ACCOUNTING DATABASE (SACCT) LOGGING & AUDIT ARCHIVES

Slurm accounting database (SACCT) logging archives full job execution records, including job ID, user, project, node count, walltime, and CPU efficiency.

Database records are archived permanently in immutable PostgreSQL database tables supporting institutional research audit requirements.

SACCT analytics generate monthly usage reports distributed to department chairs and project PIs.

Asset tracking records associate each component referenced in this section with a unique serial number and purchase order.

STRATEGY SECTION 23: HETEROGENEOUS COMPUTING NODE ALLOCATION POLICIES

Heterogeneous computing node allocation policies manage job placement across mixed node hardware architectures (CPU-only vs GPU-accelerated nodes).

Job submission filter plugins reject CPU-only workloads submitted to GPU partitions, preserving expensive GPU nodes for accelerated applications.

Node allocation rules optimize cluster resource utilization and prevent hardware misallocation.

Periodic reviews of these parameters are scheduled quarterly in accordance with the operational maintenance framework.

STRATEGY SECTION 24: REAL-TIME SLURM SCHEDULER PARAMETER AUTO-TUNING ALGORITHMS

Real-time Slurm scheduler parameter auto-tuning algorithms evaluate scheduler queue depth and dynamically adjust backfill evaluation interval times.

Auto-tuning algorithms reduce scheduler daemon CPU overhead during high-concurrency job submission spikes.

Dynamic parameter adjustments preserve scheduler responsiveness under heavy cluster loads.

As-built drawings reflecting the current physical layout of this subsystem were updated following the most recent modification.

STRATEGY SECTION 25: JOB DEPENDENCY DAG GRAPH PARSING & TOPOLOGY SORTING

Job dependency DAG graph parsing and topology sorting validate multi-job Slurm workflow dependencies prior to submission.

Dependency parsers detect circular job dependency deadlocks, alerting users before job submission.

Topology sorting optimizes job dispatch ordering across multi-stage scientific pipelines.

Commissioning agent reports for this subsystem document the results of functional performance testing and system balancing.

STRATEGY SECTION 26: HIGH-FREQUENCY JOB SUBMISSION RATE & SLURMCTLD DAEMON LOAD

High-frequency job submission rate benchmarking evaluates Slurmctld daemon response under submission bursts of up to 1,000 jobs per minute.

Daemon responsiveness tests verify zero job submission drops or RPC buffer timeouts during maximum submission rate spikes.

Daemon thread pool tuning ensures fast job dispatch times under heavy multi-user submission workloads.

Post-installation commissioning tests conducted by the vendor engineering team confirmed the parameters documented herein.

STRATEGY SECTION 27: HETEROGENEOUS COMPUTING NODE ALLOCATION POLICIES

System configuration records for Heterogeneous Computing Node Allocation Policies are maintained under version control, with each modification tracked through the institutional change management process.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Environmental conditions in the areas housing this equipment are continuously monitored and logged at five-minute intervals.

STRATEGY SECTION 28: REAL-TIME SLURM SCHEDULER PARAMETER AUTO-TUNING

Infrastructure parameters associated with Real-Time Slurm Scheduler Parameter Auto-Tuning have been validated through a combination of automated monitoring and manual inspection procedures.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.

STRATEGY SECTION 29: HIGH-FREQUENCY JOB SUBMISSION RATE DAEMON AUDIT

Operational data collected for High-Frequency Job Submission Rate Daemon Audit has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

STRATEGY SECTION 30: WORKLOAD ENERGY EFFICIENCY & JOULES-PER-JOB METRICS

The technical specifications and measured performance characteristics for Workload Energy Efficiency are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Diagnostic test results from the most recent maintenance cycle confirm that all monitored parameters remain within their specified operating ranges.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

STRATEGY SECTION 31: ANNUAL CLUSTER CAPACITY PLANNING & FY2027 EXPANSION

The technical specifications and measured performance characteristics for Annual Cluster Capacity Planning are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.